

Intuitive Surgical

NASDAQ : ISRG · Mature-Company DCF

\$350.06

\$124.3B mkt cap · 355M sh · \$9.5B cash, zero debt · 11,106 da Vinci installed base

THE CENTRAL QUESTION

Is Intuitive's near-monopoly in soft-tissue surgical robotics — 11,100 da Vinci systems, ~78% recurring revenue — durable enough to outrun the first credible competitors (Hugo, Ottawa, Versius) and justify a 50-60× forward P/FCF, or is this peak-platform pricing?

Intuitive trades at \$436.64 (\$155B mkt cap) on FY25 revenue \$10.1B, ~\$2.5B FCF, \$9.5B cash, zero debt. Structurally enviable: ~78% recurring revenue, an 11,106-system installed base growing 12% YoY, 17% procedure growth in Q1 2026, and a 26-year moat no soft-tissue competitor has broken. But the bear case is finally non-trivial: Medtronic Hugo won FDA urology clearance Dec 2025, J&J Ottawa is in trials, GLP-1s threaten bariatric volumes, and a May 2026 Class I recall compressed the multiple. Mature-Company DCF with exit-multiple terminal framing below; five scenarios, weighted; show your work.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$525.14 expected fair value today
+50.0% vs spot \$350.06

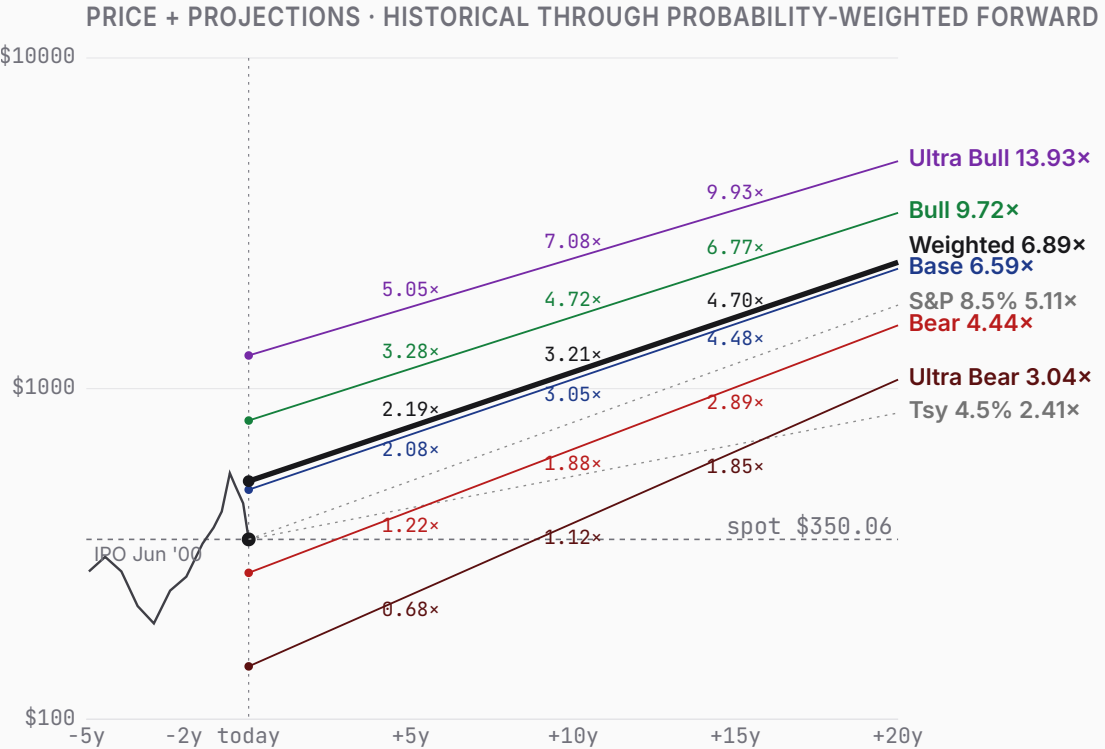
FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
\$767.63	\$1,123.07	\$1,644.63	\$2,410.70
2.19× spot	3.21× spot	4.70× spot	6.89× spot

WEIGHTING RATIONALE

- Ultra Bear 6% GLP-1s + Hugo + Ottawa + recall cycle compound; multiple compresses to ~28× FCF.
- Bear 20% Hugo lands at 8-12% of new placements; multiple resets to ~33×; revenue decelerates to 5% by FY30.
- Base 52% Mid-teens growth holds; dV5 reaccelerates utilization; recurring revenue compounds; multiple ~45×.
- Bull 16% dV5 wins general-surgery TAM; lon + cardiac add growth; multiple sustains at ~52×.
- Ultra Bull 6% Outside-OR + China + AI inflect simultaneously; multiple re-rates to ~62×.

Bull (16%) + Ultra Bull (6%) together contribute \$203.73 — 39% of the \$525.14 expected value despite 22% combined probability. Today's spot (\$350.06) sits 33% below the weighted expected; forward-weighted value crosses spot between +5y and +10y.



ULTRA BEAR	\$144.58	BEAR	\$277.34	BASE	\$494.74	BULL	\$800.86	ULTRA BULL	\$1259.91
	-58.7% vs spot		-20.8% vs spot		+41.3% vs spot		+128.8% vs spot		+259.9% vs spot
CAGR +5.2% WACC 10.5% Base 14.5K Prob 6%		CAGR +9.3% WACC 9.0% Base 16.5K Prob 20%		CAGR +12.0% WACC 8.0% Base 18.5K Prob 52%		CAGR +15.0% WACC 7.5% Base 21.0K Prob 16%		CAGR +18.8% WACC 7.0% Base 25.0K Prob 6%	
GLP-1s + Hugo + recall.		Hugo lands; multiple resets to 33×.		Mid-teens growth; razor compounds.		dV5 + general surgery wins; 52× holds.		Outside-OR + China + AI; 62× re-rate.	

PROBABILITY WEIGHTS

Ultra Bear 6% Bear 20% Base 52% Bull 16% Ultra Bull 6% · Weighted expected \$525.14 (+50.0% vs spot)

ULTRA BEAR	\$144.58	BEAR	\$277.34	BASE	\$494.74	BULL	\$800.86	ULTRA BULL	\$1259.91
	-58.7% vs spot		-20.8% vs spot		+41.3% vs spot		+128.8% vs spot		+259.9% vs spot

Probability 6%

GLP-1s + Hugo + recall.

WHY 6%

Joint conjunction of four largely independent risks: (a) GLP-1 bariatric drag at the high end (~30% volume compression); (b) Hugo + Ottawa combined share gain >15% over five years, requiring both to scale and hospitals to multi-source; (c) a quality/recall cycle that compresses the multiple structurally; (d) core-specialty (urology, GYN) procedure mix maturing earlier than expected. Each individually 15-25% conditional on macro; joint at 6%. Lower than bear's 20% because the conjunction requires multiple competitive moats to break simultaneously over a 5-year window — faster than incumbents typically lose share in regulated medtech (historical base rate: 8-12 years for a soft-tissue leader to lose 20+ points of share).

WHAT HAPPENS

The ultra-bear is a coordinated shock, not slow erosion. GLP-1 penetration compresses US bariatric volumes 30-40% over five years; Medtronic Hugo wins meaningful urology share (15-20% of new placements by FY28) and gets general-surgery clearance FY27; J&J Ottawa launches commercially FY27 and takes 5-8% of share; and a recurring quality/recall cycle (the May 2026 SureForm 30 Class I recall was the warning shot) compresses operator confidence.

Procedure growth collapses from 17% to ~6% by FY28, then 2% terminal. Installed-base growth halves from 12% to 5-6%. System ASPs compress as competitive bids force Xi-account discounting. Operating margin compresses from 29% to 22% on defensive spend. The multiple resets from ~55x to 28x FCF as the "monopoly compounder" narrative breaks. Exit-multiple SOTP: \$42B op EV + \$9.5B cash = \$52B equity + 358M shares = ~\$145/share — a -67% tail.

Probability 20%

Hugo lands; multiple resets to 33x.

WHY 20%

The competitive landscape genuinely shifted in late 2025: Hugo FDA clearance + J&J Ottawa trials + CMR Versius commercial launch. History suggests soft-tissue medtech incumbents lose 5-15 points of share over five years when the first credible competitor lands (analog: Stryker Mako vs Zimmer ROSA in ortho — Stryker held but lost some new-account placements). The 20% weight reflects that competitive entry has already happened — higher base rate than the ultra-bear — but the multiple compression from ~55x to 33x FCF is the dominant driver, more than the revenue erosion. Q1 2026 procedure growth (17%) says we're not in this scenario yet; bear is the FY27-28 thesis.

WHAT HAPPENS

The pessimistic case takes the new competitive landscape seriously. Medtronic Hugo received FDA urology clearance in December 2025 — the first credible US soft-tissue robotics competitor in 25 years. General-surgery and GYN clearances follow within 18-24 months. Hugo takes 8-12% of new system placements by FY28, primarily at price-sensitive community hospitals. J&J Ottawa enters trials and erodes the "no real competition" narrative; CMR Versius takes multi-specialty community share.

Revenue growth decelerates from 23% (Q1 2026) toward 9% by FY28 and 5% by FY30 as urology + gynecology mature and general surgery competes with Hugo on price. Operating margin holds at 27-29% — Hugo competitive-bid gross-margin pressure offset by mix toward higher-margin instruments/services on the installed base. The multiple resets from ~55x toward 33x FCF.

Probability 52%

Mid-teens growth; razor compounds.

WHY 52%

Modal outcome. Q1 2026 just printed 23% revenue and 17% procedure growth — the base case is in motion, not a future call. FY26 guidance (procedures 13.5-15.5%, gross margin 67.5-68.5%, opex 11-14%) implies a slight deceleration but no inflection. Hugo competitive entry is real but slow — 18-24 months to general-surgery clearance, 3-5 years to material share. da Vinci 5 reaccelerates installed-base utilization. The 52% weight is higher than ZM's base (50%) because (i) Q1 2026 actuals already validate the trajectory and (ii) the moat (surgeon training, switching cost, FDA data) provides more inertia than ZM's seat-based SaaS contracts.

WHAT HAPPENS

The middle path: ISRG executes on a near-monopoly that erodes slowly, not catastrophically. FY26 guidance (13.5-15.5% procedure growth) lands at the midpoint. da Vinci 5 keeps ramping — ~1,500 systems today toward ~5,000 by FY30 — driving a mid-single-digit utilization tailwind on the installed base. Ion scales (39% Q1 procedure growth annualizes toward 25-30%). Cardiac clearance opens a ~\$2B incremental specialty.

Hugo lands, but at the pace medtech rollouts actually run: 5-7% of new placements by FY28, mostly at community hospitals. Intuitive was always going to fight for. The installed-base switching cost — surgeon retraining, instrument inventory, service contracts — keeps the 11,000+ existing systems sticky. Recurring mix climbs from 78% to 81% as services and instruments compound on installed-base growth.

Probability 16%

dV5 + general surgery wins; 52x holds.

WHY 16%

da Vinci 5 captures general surgery (vs Hugo's urology-first rollout) AND ISRG holds 85%+ of incremental US placements through FY28 AND Ion scales to ~\$1.5B revenue AND the multiple sustains at ~52x FCF. The 16% weight is reasonable because (a) Q1 2026 dV5 placement velocity (232 vs 147 prior year) supports the dV5 thesis, (b) general surgery is the largest unfought TAM and ISRG has a structural advantage, but (c) sustained 50x+ multiples require continued category leadership that competitive entry directly threatens. Not the base case but a real possibility — especially if the May 2026 recall stays contained to one product line.

WHAT HAPPENS

The bull case is that da Vinci 5 unlocks the general-surgery TAM — the unfought ~70% of US soft-tissue procedures still done open or laparoscopically — AND ISRG defends share against Hugo. dV5's force feedback, on-board insufflation, and Iris imaging hit critical mass in 2027-28; placements accelerate to 2,500+/yr. Ion sustains 30%+ procedure growth toward a \$1.5B revenue line. Cardiac clearance translates to real volume. Japan, Korea, Brazil penetrate at pace.

Revenue accelerates above guidance: 17-18% CAGR through FY28, moderating to 12% by FY30. Operating margin expands toward 34% as the high-margin instrument/service razor outpaces systems. FCF reaches ~\$6.9B by FY30. The market doesn't compress the multiple because growth re-accelerates — ~52x FCF sustains. Exit-multiple SOTP: \$269B op EV + \$9.5B cash = \$279B equity + 348M shares = ~\$801/share. +83% from spot.

Probability 6%

Outside-OR + China + AI; 62x re-rate.

WHY 6%

Joint conditional: outside-OR / ASC deployment scales (40% conditional on a dV5 compact form factor) AND China crosses \$2B by FY30 (30% conditional on no geopolitical catastrophe) AND AI / autonomous-surgery monetization gets real (25% conditional on an FDA approval pathway for semi-autonomous assist) AND the multiple re-rates to 60-65x FCF (40%). Joint at 5-7%, reflected at 6%. Tail of tails — but ISRG has the installed base, the data moat, and the cash to actually attempt all three simultaneously.

WHAT HAPPENS

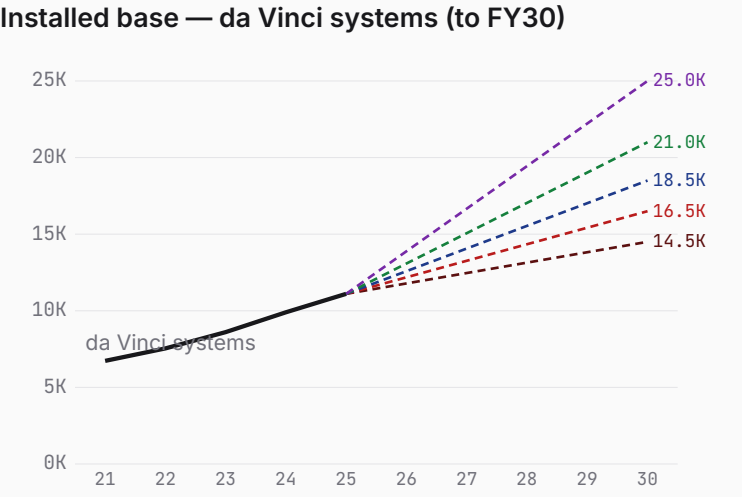
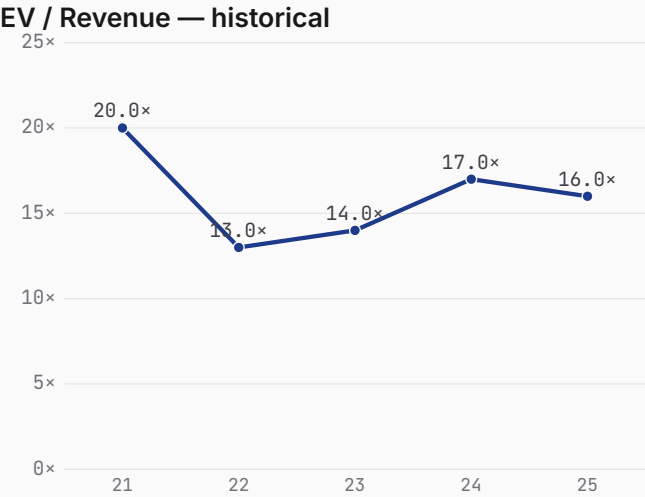
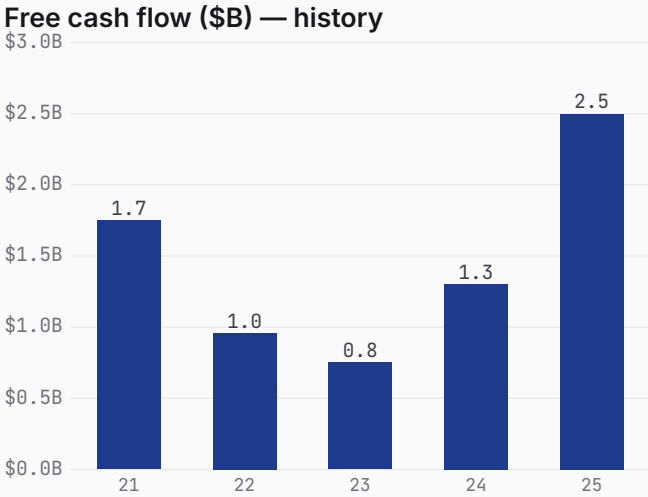
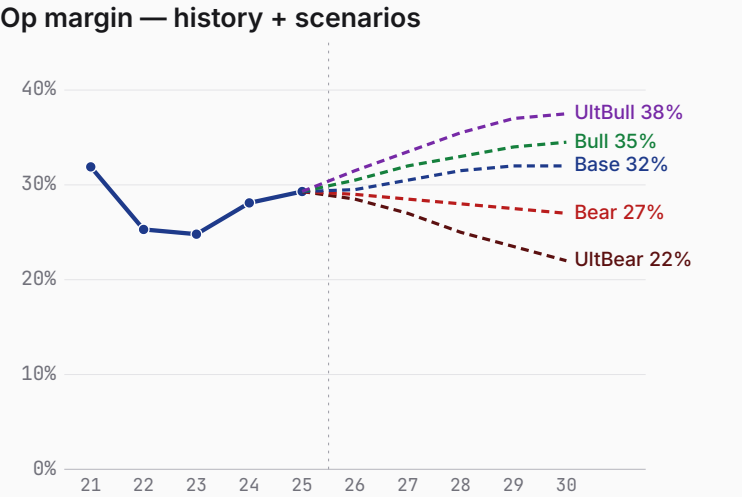
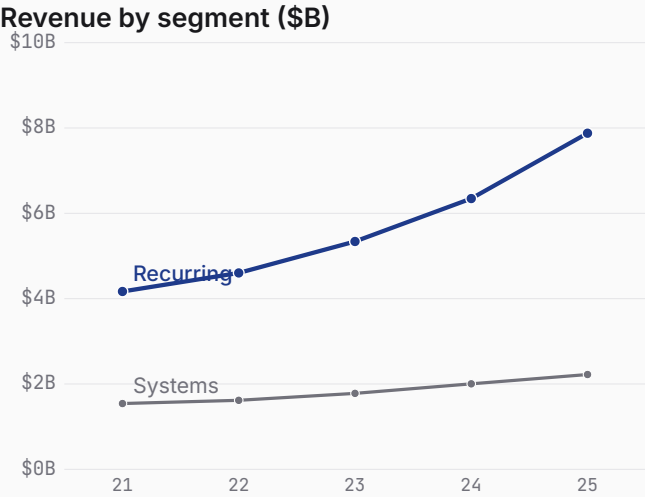
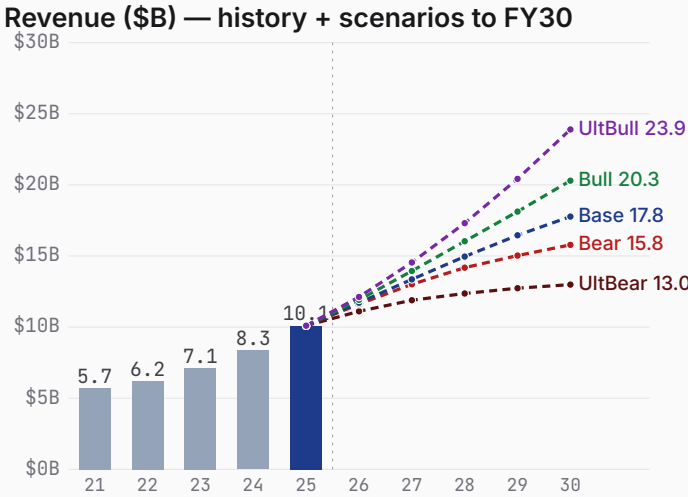
The ultra-bull is ISRG inflecting on three independent axes at once. (1) Outside-OR / clinic deployment: dV5 compact configurations enable ambulatory-surgical-center placements at \$500K-1M (vs \$2.5M traditional), tripling the addressable TAM beyond hospital ORs. (2) China + India scale: China revenue grows from <\$500M to >\$3B by FY30 as the Fosun JV penetrates tier-2/3 cities; India localized manufacturing enables local-market price points. (3) AI / autonomous surgery: ISRG's data moat (17M+ recorded procedures) becomes training data for semi-autonomous assistance — a new SaaS-like recurring line.

Revenue compounds at ~19% CAGR — \$10.1B → \$25B+ by FY30. Operating margin reaches 36-37% as 90%+ gross-margin software/AI blends in. FCF: \$2.5B → ~\$9B. The market re-rates to growth-software multiples: ~62x FCF on a \$9B base. Exit-multiple SOTP: \$425B op EV + \$9.5B cash = \$435B equity + 345M shares = ~\$1,260/share. +189% from spot. This is the scenario where ISRG stops being "great mature medtech" and becomes "the SaaS-like platform for procedures."

Intuitive Surgical business snapshot

Bear \$277.34 Base \$494.74 Bull \$800.86

FY21–FY25 history + FY26–FY30 scenario projections · fiscal years end Dec 31 · 10-K FY25, Q1 2026 10-Q, FDA filings, medtech competitive analysis



Sources: ISRG 10-K FY25, Q1 2026 8-K/10-Q, Medtronic Hugo FDA clearance Dec 2025, J&J Ottava trial filings, surgical-robotics TAM analysis.

Intuitive Surgical 7 Powers · the moat, scored

Bear \$277.34 Base \$494.74 Bull \$800.86

Arena. Soft-tissue robotic surgery — da Vinci's installed base vs. emerging challengers (Medtronic Hugo, J&J Ottava, CMR Surgical Versius) for the global multi-specialty minimally-invasive surgery platform standard.

POWER AUDIT
dominant-power durability: high

POWER AUDIT · 7 POWERS

Scale Economies 	~9,500 systems installed; instrument/service volume amortizes R&D no challenger matches.
Network Economies 	Surgeon training + case-data flywheel (Case Insights) compounds with the installed base.
Counter-Positioning 	Challengers catch up on model, not displace it; weak collateral-damage barrier.
Switching Costs · thesis leans here 	Sunk capital + surgeon retraining + hospital workflow lock-in; razor/blade recurring revenue.
Branding 	da Vinci is the category synonym; surgeon and hospital default.
Cornered Resource 	Deep patent estate (many now expiring) + the installed base itself as a distribution asset.
Process Power 	Two decades of manufacturing/service learning-curve and reliability data.

COMPETITIVE READ

A wide, durable moat anchored on switching costs and scale — the basis for the DCF's premium terminal multiple; challengers are real but years from the installed-base and brand barriers. Watch procedure growth as the erosion gauge.

PEER SET

Medtronic (Hugo) CE-marked; US trials; bundled into a broad device channel.	10% gr · 25% mgn · ~17x EV/EBIT
J&J MedTech (Ottava) Delayed; deep surgical channel + Ethicon stapling tie-in.	6% gr · 28% mgn · ~16x EV/EBIT
CMR Surgical (Versius) Modular, lower-cost; OUS traction.	30% gr · private

THREAT VECTORS · FALSIFIERS

Counter-Positioning · Medtronic / J&J falsifier: ISRG procedure growth < 10% for 2 consecutive years while a rival's installed base passes 1,000 systems.
Switching Costs · Hospitals (capital pressure) falsifier: da Vinci placement mix shifts to lease/usage > 50%, signaling pricing-power erosion.

Intuitive Surgical show your work

Bear \$277.34 Base \$494.74 Bull \$800.86 · Weighted \$525.14 (+50.0%)

ULTRA BEAR					BEAR					BASE					BULL					ULTRA BULL				
\$144.58					\$277.34					\$494.74					\$800.86					\$1259.91				
ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS				
5y revenue CAGR (%)					+5.2					5y revenue CAGR (%)					+9.3					5y revenue CAGR (%)				
Year-5 op margin (%)					22.0					Year-5 op margin (%)					27.0					Year-5 op margin (%)				
WACC (%)					10.5					WACC (%)					9.0					WACC (%)				
Exit FCF multiple (x)					28					Exit FCF multiple (x)					33					Exit FCF multiple (x)				
Installed base FY30					14,500					Installed base FY30					16,500					Installed base FY30				
Starting revenue (\$B)					10.10					Starting revenue (\$B)					10.10					Starting revenue (\$B)				
Scenario probability (%)					6					Scenario probability (%)					20					Scenario probability (%)				
5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B				
FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF
FY26	11.11	+28%	+2.50	+2.26	FY26	11.61	+29%	+2.85	+2.62	FY26	11.72	+30%	+3.20	+2.96	FY26	11.92	+31%	+3.65	+3.40	FY26	12.12	+32%	+3.82	+3.57
FY27	11.89	+27%	+2.45	+2.01	FY27	13.01	+28%	+3.13	+2.63	FY27	13.36	+31%	+3.70	+3.17	FY27	13.94	+32%	+4.55	+3.94	FY27	14.54	+34%	+4.88	+4.26
FY28	12.36	+25%	+2.30	+1.71	FY28	14.18	+28%	+3.32	+2.56	FY28	14.96	+32%	+4.20	+3.33	FY28	16.04	+33%	+5.45	+4.39	FY28	17.31	+36%	+6.20	+5.06
FY29	12.73	+24%	+2.13	+1.43	FY29	15.03	+28%	+3.40	+2.41	FY29	16.45	+32%	+4.55	+3.34	FY29	18.12	+34%	+6.15	+4.61	FY29	20.42	+37%	+7.55	+5.76
FY30	12.99	+22%	+1.98	+1.20	FY30	15.78	+27%	+3.50	+2.27	FY30	17.77	+32%	+4.85	+3.30	FY30	20.29	+35%	+6.85	+4.77	FY30	23.89	+38%	+9.05	+6.45
Σ PV explicit FCF					+8.61					Σ PV explicit FCF					+21.09					Σ PV explicit FCF				
+ PV terminal (g 2.0%)					33.65					+ PV terminal (g 3.0%)					248.11					+ PV terminal (g 4.0%)				
EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE				
Operating EV					\$42.26B					Operating EV					\$269.20B					Operating EV				
+ Cash & investments					\$9.50B					+ Cash & investments					\$9.50B					+ Cash & investments				
= Equity value					\$51.76B					= Equity value					\$278.70B					= Equity value				
FY30 shares (M)					358					FY30 shares (M)					348					FY30 shares (M)				
= Expected per share					\$144.58					= Expected per share					\$800.86					= Expected per share				
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT				
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y	
\$238.19	\$392.40	\$646.46	\$1,065.01		\$426.72	\$656.56	\$1,010.21	\$1,554.33		\$726.94	\$1,068.11	\$1,569.40	\$2,305.96		\$1,149.74	\$1,650.60	\$2,369.65	\$3,401.93		\$1,767.09	\$2,478.43	\$3,476.13	\$4,875.45	
0.68×	1.12×	1.85×	3.04×		1.22×	1.88×	2.89×	4.44×		2.08×	3.05×	4.48×	6.59×		3.28×	4.72×	6.77×	9.72×		5.05×	7.08×	9.93×	13.93×	



Intuitive Surgical People · Opportunity · Context · Deal

Bear \$277.34 Base \$494.74 Bull \$800.86

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Dave Rosa (CEO since Jul 2025)

1%insider ownership

LOWkey-person risk

Capital allocation (realized)
Net cash, zero debt, no dividend — a self-funded compounder: reinvests first (R&D +~15% YoY 2024 + manufacturing capex), buys back opportunistically (~\$7.95B cumulative, \$5B authorization refreshed), and grows organically (build-not-buy, no transformative M&A).

Incentive alignment
Pay is ~85-89% equity, performance-weighted: PSUs pay 0-125% of target over 3 years on relative TSR + da Vinci/Ion procedure-volume growth + relative operating margin. Guthart FY2024 ~\$18.2M (mostly stock, \$0 options).

GOVERNANCE FLAGS

- single-class (one share, one vote); no controlling individual (insiders <1%, ~85% institutional)
- declassified board — directors elected annually; fully-independent committees; say-on-pay ~93% (2025)
- orderly succession: Rosa is a 29-yr internal promotion; Guthart stayed on as Executive Chair
- caveat: with Guthart as Executive Chair the chair is no longer independent (lead independent director)

PEOPLE READ

The cleanest governance of the cohort: single-class, declassified annual elections, ~93% say-on-pay, performance-weighted equity, net-cash discipline, and an orderly internal CEO handoff (Rosa) — institutionalized, not person-dependent. Mild demerits: low insider skin-in-the-game and a now non-independent (Executive) chair.

THE FOUR LEGS

People observable composite · 1–5

Opportunity
The scenario distribution · the Page-3 business snapshot · the Arthur Indicator (ISRG 17.5 — rarely screens cheap)

Context
The 7 Powers analysis (Power Audit)

Deal
Open-market purchase = the deal: entry price vs. the scenario distribution (the +20% finding) + position sizing.



Intuitive Surgical

supporting analysis · disclaimers · glossary

Bear \$277.34 Base \$494.74 Bull \$800.86

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1 **Moat is real and measurable.**
11,106 installed base + 17M procedures + 26-year FDA record. Hugo has <50 US systems, one specialty cleared. The asymmetry is enormous.
- 2 **Razor-blade is the business.**
78% of FY25 revenue recurring (instruments + service). Compounds on installed-base growth even if system pricing compresses.
- 3 **Balance sheet is a fortress.**
\$9.5B cash, zero debt, \$4B buyback authorization. Q1 2026 retired \$1.1B of stock. Capital return finally at scale.
- 4 **da Vinci 5 is reaccelerating.**
Q1 2026: 232 dV5 placements vs 147 prior year (+58%). US utilization +4%. Force feedback is a step-function, not incremental.
- 5 **Procedure growth is structural.**
17% Q1 procedure growth on already-trained surgeons. Grows with aging + minimally-invasive penetration. Hard to bend the curve.

FALSIFICATION TRIGGERS

Bear validation Hugo wins > 15% of US new placements · dV5 placements < 200/qtr · procedure growth < 10% · another Class I recall on a core line · forward P/FCF < 30×	Bull validation dV5 placements > 300/qtr · Ion procedure growth > 40% · cardiac volume > 5,000/yr · general-surgery share grows materially · forward P/FCF > 50×	Reframe needed If Stryker / J&J announce a credible cross-platform alliance against ISRG, OR if ISRG acquires a major outside-OR or AI surgical asset (>\$5B deal), the moat / TAM framing changes materially
--	--	---

DISCLAIMERS · PLEASE READ BEFORE USING THIS DOCUMENT

Not investment advice. This document is produced for entertainment and educational purposes only by an individual amateur investor. "Alameda Research 2: Electric Boogaloo" is the unincorporated personal concept of one person and is not a registered investment management entity. Nothing herein constitutes investment advice, a recommendation, or a solicitation to buy or sell any security.	Not a professional. The author is not a registered investment advisor, broker-dealer, CFA charterholder, certified financial planner, or licensed financial professional of any kind. The author has no formal training in equity research or capital markets. No fiduciary relationship is created by reading this document.	AI-assisted analysis. Substantial portions of this analysis were produced with the assistance of large language models. LLMs are known to fabricate facts, miscalculate numbers, hallucinate sources, and present incorrect information with high apparent confidence. Every figure, claim, and projection in this document should be independently verified before acting on it.
Forward-looking statements. Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.	Author may hold positions. The author may own, intend to acquire, or hold short exposure to \$ISRG or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.	Do your own research. Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

da Vinci 5 — ISRG's 5th-gen surgical robot (2024). Force feedback, on-board insufflation, Iris imaging. ~1,500 installed; driving +4% US utilization.	Ion — ISRG's endoluminal robot for minimally invasive lung biopsies. Q1 2026 procedure growth +39% YoY. Separate installed base.	Hugo (Medtronic) — Medtronic's competing soft-tissue robot. FDA-cleared Dec 2025 for urology — first credible US soft-tissue competitor in 25 years.
Ottava (J&J) — J&J's competing soft-tissue robot. In trials FY26; commercial launch expected FY27-28. Adjacency to J&J's Monarch.	Installed base — da Vinci systems in service worldwide. 11,106 at Dec 2025 (+12% YoY). The denominator that drives recurring-revenue compounding.	Recurring revenue mix — Revenue from instruments, accessories, and service (vs one-time systems). FY25 ~78%. The razor-blade lock-in.